

Knowing One’s Self is Wisdom: An Analysis of an LLM’s Knowledge of Its Own Decoder

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Project write-up, *Reliable ML with Unreliable Black Boxes*

Abstract

An autoregressive language model is trained to *predict* text, so when it *generates* it has an incentive to model its own *decoder* — the temperature, top- k cutoff, or banned-token list that turns its logits into tokens. Does the model know that decoder, and can we read it off the model’s own logits, training-free? We report preliminary steps, in three beats. **(1)** The model tracks its decoder: ban a token or impose a top- k cutoff while it generates, and the model — reading back its own constrained text — lowers the right probabilities. **(2)** A streaming estimator reads an effective inverse-temperature $\beta = 1/T$ off the prefix one token at a time; on known-temperature self-text it recovers the true $1/T$ and predicts near-optimally, though natural text carries little to recalibrate (Qwen-2.5-3B is temperature-robust). **(3)** Our central result is a **partial proof**: building that estimator as a grid (“bucket”) of fixed-temperature experts makes it exactly Vovk’s Aggregating Algorithm for log loss, and it therefore carries two *per-sequence, distribution-free* guarantees on arbitrary streams — a forecast-regret bound of $\ln G$ against the best bucket, and a convex level-set error bar around the grid MLE (Lemma 1) — while certifying the Beat-2 read-out. Turning the read-out into better *generation* self-neutralizes and is left open. All numbers are script-generated and cited by name.

1 Does the model know its own decoder?

At each step an autoregressive language model emits a distribution over next tokens; *decoding* chooses how to sample from it. Because models are trained to predict text rather than to generate it, a model that reads back its own generated prefix has an incentive to infer the decoder that produced it and adjust its predictions accordingly (Basu et al., 2020). The question this project addresses is whether that decoder — concretely, an effective inverse-temperature $\beta = 1/T$ — can be *read off the model’s own logits online, training-free, and with guarantees*, and whether the read-out is useful for prediction or generation. This sits at the intersection of post-hoc calibration (Guo et al., 2017; Liu et al., 2023) and the algorithms-with-predictions viewpoint (Mitzenmacher and Vassilvitskii, 2020) of the course, in which a possibly-unreliable prediction augments a downstream decision; what we then most want is a *worst-case, per-sequence* guarantee that holds even when the stream is misspecified, which Section 3 delivers.

The model tracks its decoder. We constrain the decoder while the model generates, then ask whether the model, reading back its own constrained text, has inferred the constraint. **Blacklist:** ban a token during generation; the model lowers its predicted probability on that token across the sequence ($\Delta p_{\text{late-early}} = -0.019$ for the top token, and below an unbanned control -0.016 , 95% CI $[-0.019, -0.013]$), with an early-position contrast ≈ 0 as a sanity check (Figure 1). **Top- k :**

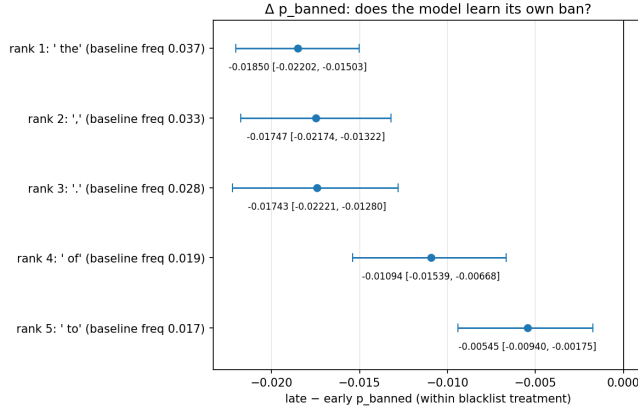


Figure 1: Beat 1. With one token banned during generation, the model’s predicted probability on the banned token falls across the sequence (late minus early, $\Delta p = -0.019$ for the top token) and below an unbanned control — evidence the model has inferred the blacklist from its own text.

truncating to the top k produces no cliff at k itself — probability instead concentrates on the head, more so the tighter the cutoff. So the model tracks its decoder, though the top- k signal is broad sharpening, not a k -specific edge.

2 A streaming estimator for the effective temperature

If the model tracks its decoder, can we make that quantitative and *useful*? A streaming *filter* reads an inverse-temperature $\hat{\beta}_t = 1/\hat{T}_t$ off the prefix one token at a time, training-free, using only the model’s own logits.¹ **Recovery:** on Qwen self-text generated at a *known* temperature (64 sequences per T), $\hat{\beta}_t$ settles on the true $\log(1/T)$ from the prefix alone (bias -0.030 at $T=0.7$; Figure 2, left). **Prediction:** using $\hat{\beta}_t$ as a per-token temperature beats the default $\beta=1$ model by 0.108 nats at $T=0.7$ (up to 0.723 at $T=2$), within $+0.025$ nats of an oracle that *knows* $1/T$ (Figure 2, right). Because the filter uses only information the model itself has, it is something the model could in principle run internally to predict better. *Natural* text, by contrast, carries little to recalibrate: fitting the best global temperature across 9 types from in-distribution to far-OOD, Qwen is temperature-robust everywhere — $T^* \in [0.91, 1.08]$, gain $\leq +0.0032$ nats on any real type ($+0.0283$ only on shuffled gibberish).

What the filter lacks. This filter is fast, but its experts are *not* fixed (it re-linearizes the likelihood at $\beta=1$ each step), so it carries *no* per-sequence guarantee: on an adversarial or misspecified stream nothing bounds how wrong $\hat{\beta}_t$ can be. Supplying exactly that guarantee — and certifying this filter’s outputs against it — is the subject of the next section.

¹A prior-regularized one-step online MLE on $\eta = \log \beta$: with a log-normal prior (mean 1, scale σ_0) and a per-token Newton/Fisher step of the categorical likelihood under $q = \text{softmax}(\ell_t)$, using $\text{score}_t = \ell_t[x_t] - \mathbb{E}_q[\ell_t]$ and $J_t = \sigma_0^{-2} + \sum_{\tau \leq t} \text{Var}_q[\ell_\tau]$ evaluated at $\beta=1$. Implementation and unit tests (shift invariance, batched=serial, recovery under matched-prior synthetic data): `src/decoding_decoding/counter_decode.py`.

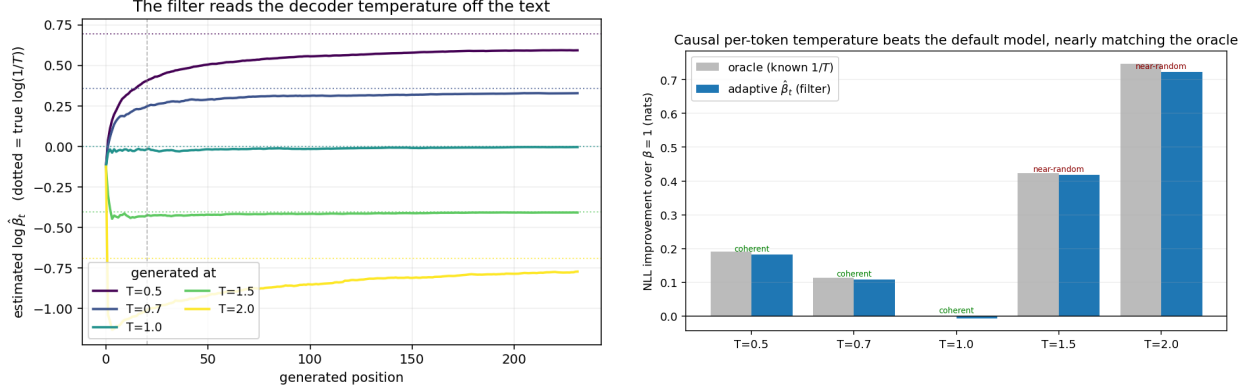


Figure 2: Beat 2. *Left*: on known-temperature self-text the streaming $\hat{\beta}_t$ recovers the true $\log(1/T)$ from the prefix alone. *Right*: using $\hat{\beta}_t$ as a per-token temperature lowers held-out NLL toward the oracle that knows $1/T$. (Large $T \geq 1.5$ gains are on near-random text; coherent wins are at $T=0.5, 0.7$.)

3 A streaming estimator *with guarantees*: the bucket certificate

Fix a grid (a set of “buckets”) of inverse-temperatures $\beta_1 < \dots < \beta_G$, log-spaced over a bounded range. Treat each β_g as a fixed *expert* that forecasts the next token with probability $P_g(x_t) = \text{softmax}(\beta_g \ell_t)[x_t]$ on the model’s *raw* logits $\ell_t \in \mathbb{R}^V$ (no rescaling — fixed experts are what make the guarantees hold). Let

$$L_n(\beta) = \sum_{t \leq n} [\text{lse}(\beta \ell_t) - \beta \ell_t[x_t]], \quad \text{lse}(\cdot) = \log \sum_v \exp(\cdot),$$

be the cumulative log loss of bucket β , let π_0 be a prior on the grid, and let the streaming forecaster predict x_t with probability $\sum_g w_g(t) P_g(x_t)$, where $w_g(t) \propto \pi_0[g] e^{-L_{t-1}(\beta_g)}$ is the Bayes posterior over buckets. This is precisely Vovk’s Aggregating Algorithm for log loss (Vovk, 1998; Cesa-Bianchi and Lugosi, 2006), and the following holds for *every* logit/token sequence — with no assumption that tokens were sampled from any P_β , or from anything at all.

Lemma 1 (per-sequence certificate). *For any logit sequence $\ell_{1:n}$ and any token sequence $x_{1:n}$: (i) the forecaster’s cumulative log loss is $-\ln \sum_g \pi_0[g] e^{-L_n(\beta_g)} \leq L_n(\beta_g) + \ln(1/\pi_0[g])$ for every g ; with uniform π_0 the regret against the best bucket in hindsight is at most $\ln G$. (ii) With uniform π_0 the posterior mode equals $\arg \min_g L_n(\beta_g)$, the best-fit (effective) inverse temperature on the grid, exactly. (iii) L_n is convex with $L_n''(\beta) = \sum_t \text{Var}_{q_{\beta,t}}[\ell_t]$, $q_{\beta,t} = \text{softmax}(\beta \ell_t)$, so for any $c > 0$ the sub-level set $\{\beta : L_n(\beta) \leq \min L_n + c\}$ is an interval containing the sequence MLE; its endpoints, read off the stored grid loss profile, are a data-dependent error bar that needs no distributional assumptions.*

Proof. (i) The forecaster’s per-step log probabilities telescope:

$$\sum_t \ln \sum_g w_g(t) P_g(x_t) = \ln \sum_g \pi_0[g] e^{-L_n(\beta_g)} \geq \ln (\pi_0[g] e^{-L_n(\beta_g)}) \text{ for each } g.$$

(ii) Immediate from $w_g(n) \propto e^{-L_n(\beta_g)}$. (iii) Each summand of L_n is an lse of an affine function of β minus an affine function, hence convex; differentiate twice. $\square$

pre-registered hypothesis	outcome	test	verdict
H1 coverage (synthetic, well-spec.)	200/200 cover the MLE	$p = 1.00$	passes
H2 non-vacuousness (Qwen, $\beta=1$)	median width 0.05 (< 0.2 in $\log \beta$)	$p < 10^{-9}$	holds
H3 degradation (peaked vs. natural)	median 0.45 vs. 0.25	$p < 10^{-67}$	holds
H4 transfer (faithful Laplace in bar)	88/96 inside	$p = 0.76$	passes

Table 1: Pre-registered evaluation of the certificate (hypotheses, sample sizes, and thresholds committed before any data were generated). Real streams are Qwen-2.5-3B, 32 prompts, 1000 tokens.

“Distribution-free” here does double duty. Over the data process, the lemma assumes nothing about how $x_{1:n}$ was produced. Over the estimator’s own assumptions, with uniform π_0 the mode in (ii) *is* the grid MLE and the bar in (iii) is an exact profile-likelihood interval (not a Gaussian approximation); the prior survives only as the constant $c = \ln G$. We report the mode, not the posterior mean: by (ii) it equals the best-fit bucket with zero slack, whereas the mean of the (unimodal but skewed — e^{-L_n} is log-concave by (iii)) posterior over a log-spaced grid is mildly biased.

Three honest caveats. First, the certificate is relative to the in-hindsight best-fit β , not the dial a counterparty actually set; under a misspecified decoder (a top- k cutoff, a blacklist) no single “true” β describes the stream, and the best-fit value is exactly what “effective temperature” should mean there. The bar then certifies *fit to the observed stream*, and a narrow bar means β is *identifiable*, not that any temperature fits well. Second, the interval in (iii) widens exactly when peaked logits make β unidentifiable ($\text{Var}_q[\ell_t] \rightarrow 0$), so forecasting regret and estimation precision degrade in opposite directions. Third, the guarantee attaches to the mixture, not a point estimator (proper online predictors face an exp-concavity obstruction that improper mixtures avoid (Foster et al., 2018)), so transferring it to the fast filter (Section 2) is an empirical, pre-registered claim.

Pre-registered validation. We committed four hypotheses, with sample sizes and thresholds, *before* generating any data (`paper/grid/certificate/prereg.md`); the bar uses $c = \ln G$ on a 121-bucket grid spanning $\log \beta \in [-3, 3]$. Table 1 reports the outcome. The bar covered the continuum MLE on every well-specified synthetic stream (200/200); on real Qwen-2.5-3B streams at $\beta=1$ the median width was 0.05 in $\log \beta$ (the grid-resolution floor — the bar is far from vacuous where logits carry temperature information) and widened on peaked-logit streams exactly as (iii) predicts. The transfer check to the Beat-2 filter passed (88/96 of estimates inside the bar), while the original $\beta=1$ -linearized filter landed inside on only 10/96 — so the certificate extends to the corrected filter’s outputs, not the original’s.

4 Generating better, and limitations

Can we use the read-out to *generate* better? Not yet. The tempting move is to feed $\hat{\beta}_t$ back into the sampler to “correct” the temperature as it generates. It self-neutralizes: correcting toward $\hat{\beta}$ drives the effective decoding temperature back to $\beta_{\text{dec}} \rightarrow 1$, quietly cancelling the very signal it reads off. We confirmed this across exact-grid filters, so it is not an approximation artifact (Appendix A). Turning the read-out into better generation is the honest open problem.

Limitations. These are preliminary steps, on a single model and tokenizer; the certificate certifies the effective β of the *observed* stream, never a counterparty’s dial. The natural next step, for which it is the right tool, is misspecification: the *effective* temperature and goodness-of-fit that top- k and blacklist decoders register, where no single temperature reproduces the stream and the best-fit bucket is a projection (synthetic pilot: `scripts/e1_synthetic_pilot.py`).

A Corrector and supporting control experiments

A corrector $\beta_t^{\text{dec}} = \beta^{\text{target}} / \hat{\beta}_t^p$ divides a user’s target by the estimate. An empirical sweep shows it pins the effective temperature in the “boredom” regime ($\beta^{\text{target}} > 1$) but overshoots in the “confusion” regime ($\beta^{\text{target}} < 1$), where low- β self-generation degrades into incoherent text within ~ 20 tokens (a garbage-in/garbage-out confound that limits what the confusion-regime numbers can claim). Two further controls in the longer draft (`paper/decoding_paper.tex`) — a sampler-switch experiment (the belief is not a fast per-token entropy read-out, a null on one mechanism) and a closed-loop-vs-invariance comparison — support only “do not correct at $\beta^{\text{target}} = 1$, because Qwen is already calibrated there.” The natural-text survey behind Beat 2’s wild-text claim (all 9 types) is tabulated there as well.

References

- Sourya Basu, Govardana Sachitanandam Ramachandran, Nitish Shirish Keskar, and Lav R. Varshney. Mirostat: A neural text decoding algorithm that directly controls perplexity, 2020. URL <http://arxiv.org/abs/2007.14966v2>.
- Nicolo Cesa-Bianchi and Gabor Lugosi. *Prediction, Learning, and Games*. Cambridge University Press, March 2006. ISBN 9780511546921. doi: 10.1017/cbo9780511546921. URL <http://dx.doi.org/10.1017/CB09780511546921>.
- Dylan J. Foster, Satyen Kale, Haipeng Luo, Mehryar Mohri, and Karthik Sridharan. Logistic regression: The importance of being improper, 2018. URL <http://arxiv.org/abs/1803.09349v2>.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks, 2017. URL <http://arxiv.org/abs/1706.04599v2>.
- Tong Liu, Iza Škrjanec, and Vera Demberg. Temperature-scaling surprisal estimates improve fit to human reading times – but does it do so for the “right reasons”?, 2023. URL <http://arxiv.org/abs/2311.09325v2>.
- Michael Mitzenmacher and Sergei Vassilvitskii. Algorithms with predictions, 2020. URL <http://arxiv.org/abs/2006.09123v1>.
- V Vovk. A game of prediction with expert advice. *Journal of Computer and System Sciences*, 56(2):153–173, April 1998. ISSN 0022-0000. doi: 10.1006/jcss.1997.1556. URL <http://dx.doi.org/10.1006/jcss.1997.1556>.